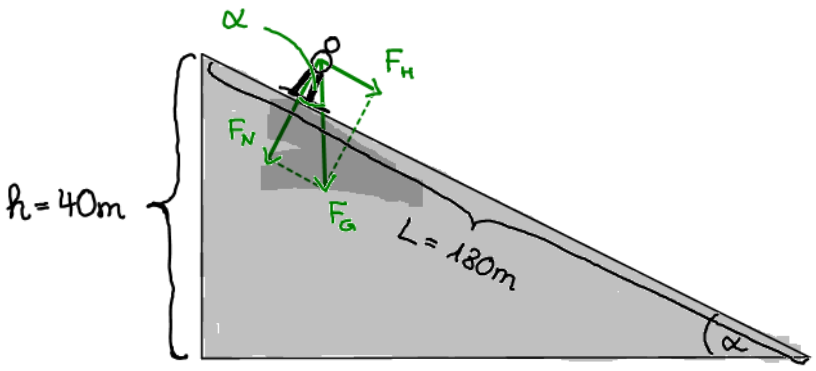
Lösung: Skiläufer und Reibung



(i) Winkel α : (aus Geometrie des Abhangs)

$$\sin(\alpha) = \frac{h}{L}$$

(ii) Coulomb-Reibung: $F_R = -\mu \cdot F_N = -\mu \cdot \cos(\alpha) \cdot F_G$

(iii) Hangabtriebskraft $F_H = \sin(\alpha) \cdot F_G$

(iv) Beschleunigung des Skilaufers

$$a_S = \frac{F_{ges}}{m_S} = \frac{F_H + F_R}{m_S} = \frac{\sin(\alpha) \cdot F_G - \mu \cdot \cos(\alpha) \cdot F_G}{m_S}$$

Gesamtkraft $F_{ges} = F_H + F_R$

$$a_S = \frac{F_G}{m_S}\left(\sin(\alpha) - \mu \cdot \cos(\alpha)\right) \qquad \text{mit} \quad F_G = m \cdot g$$

$$\underline{\underline{a_S = g \cdot \left(\sin(\alpha) - \mu \cdot \cos(\alpha)\right)}}$$

(v) Weg-Geschwindigkeits-Gesetz (gleichmäßig beschl. Bew.)

$$\left.\begin{aligned} L &= \tfrac{1}{2}\, a_S \cdot t_F^2 \\ v_{end} &= a_S \cdot t_F \Leftrightarrow t_F = \frac{v}{a_S} \end{aligned}\right\} \Rightarrow \underline{\underline{L = \frac{v_{end}^2}{2a_S}}}$$

(vi) Reibungskoeffizient μ

$$\left.\begin{aligned} a_s &= \frac{v_{end}^2}{2L} \\ a_s &= g\cdot\left(\sin(\alpha)-\mu\cdot\cos(\alpha)\right) \end{aligned}\right\} \qquad \frac{v_{end}^2}{2L} = g\cdot\left(\sin(\alpha)-\mu\cdot\cos(\alpha)\right)$$

$$\Rightarrow \quad \frac{v_{end}^2}{2gL} = \sin(\alpha)-\mu\cdot\cos(\alpha)$$

$$\Rightarrow \quad \sin(\alpha)-\frac{v_{end}^2}{2gL} = \mu\cdot\cos(\alpha)$$

$$\Rightarrow \quad \mu = \frac{\sin(\alpha)}{\cos(\alpha)} - \frac{v_{end}^2}{2gL\cdot\cos(\alpha)}$$

$$= \frac{\sin(\alpha)}{\cos(\alpha)}\left(1-\frac{v_{end}^2}{2gL\sin(\alpha)}\right) \qquad \text{mit} \quad \cos(\alpha)=\sqrt{1-\sin^2(\alpha)}$$

$$= \frac{\sin(\alpha)}{\sqrt{1-\sin^2(\alpha)}}\left(1-\frac{v_{end}^2}{2gL\sin(\alpha)}\right) \qquad \text{mit} \quad \sin(\alpha)=\frac{h}{L}$$

$$= \frac{h/L}{\sqrt{1-(h/L)^2}}\left(1-\frac{v_{end}^2}{2gh}\right)$$

$$\mu = \frac{h}{\sqrt{L^2-h^2}}\left(1-\frac{v_{end}^2}{2gh}\right)$$

entspricht $72\,\frac{km}{h}$

$$\mu \approx \frac{40m}{\sqrt{(180m)^2-(40m)^2}}\left(1-\frac{(20\,m/s)^2}{2\cdot 9{,}81\,m/s^2\cdot 40m}\right) \approx 0{,}11$$

Einheitenlos!!!